Toward Architecture-Driven Interdisciplinary Research

Learnings from a Case Study of COVID-19 Contact Tracing Apps

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ABSTRACT

This paper explores the use of an architectural perspective to study complex data ecosystems and to facilitate a normative discourse on such ecosystems. It argues that an architectural perspective is helpful to bridging discursive and methodological gaps between information systems (IS) research and legal studies. Combining architectural and normative perspectives is a novel interdisciplinary research approach that provides a framework for analyzing techno-legal contexts. The merits and challenges of this approach are demonstrated and discussed in this paper using the example of COVID-19 contact tracing apps. We conceptualize our results on three levels of knowledge: the first is the actual knowledge of the exemplary contact tracing app we studied and its ecosystem; the second is knowledge of the architectural meta-model that we used, its benefits and its shortcomings; and the third is knowledge of the interdisciplinary research process of acquiring common knowledge shared by IS scholars and legal experts.

CCS CONCEPTS

• **Applied computing** → *Enterprise architectures; Law, social and behavioral sciences;* • **Security and privacy** → *Human and societal aspects of security and privacy*

KEYWORDS

Architecture, contact tracing app, COVID-19, data ecosystem, HaMagen, information systems, interdisciplinary research, law, privacy

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1 INTRODUCTION

This paper describes the explorative process of a research team consisting of information systems (IS) and legal scholars. The team explored the use of an architectural perspective to study complex data ecosystems and used this perspective to facilitate a normative discourse on such ecosystems, with the aim of bridging discursive and methodological gaps between IS research and legal studies. Combining architectural and normative perspectives is a novel interdisciplinary research approach that offers a framework for analyzing techno-legal contexts, as will be unfolded in the following. We used COVID-19 contact tracing apps as an example for this exploration.

Since the outbreak of the global pandemic, COVID-19 digital surveillance has been increasingly deployed by governments around the world to track and notify individuals about contact with confirmed COVID-19 patients (e.g., Google and Apple interfaces or the PEPP-PT approach); to enforce mandatory self-isolation (e.g., Hong Kong); to automate clearance for employees in the job market; to allowing entry into businesses or access to public transportation (e.g., China); or to predict the likelihood of an outbreak in certain areas [1, 5]. Common to all these measures is the collection and analysis of sensitive health, location, and proximity data at various levels of precision. The use of these measures has raised a vivid public debate over the appropriate balance, regarding the design of a particular system's architecture, between public health necessities and fundamental human rights, especially in the context of privacy and data protection [1, 31].

However, focusing exclusively on the architecture of a specific technology or app might be too narrow and may suffer from a blind spot. It risks overlooking the ways by which these newly

introduced measures are integrated into an existing technological, economic, and political ecosystem where different players may access data and exercise economic or political power, and data might be used in different contexts and for different purposes. Consequently, the risks and benefits of any such measures should not be analyzed in isolation, but rather should be understood in the context of complex data ecosystems wherein incentives, affordances, and costs are shaped by different players and different architectures [7, 25].

Different architectures have already emerged for COVID-19 tracking and mitigation and are deployed and used by different actors in different environments [1]. While the first waves of COVID-19 contact tracing apps have already been deployed and researchers have been involved in their development through critical review [1, 5], this project aims at laying the groundwork for a debate about the inevitable evolvement and potentially expanded use of those technologies as well as for their critical assessment regarding potential readjustments. We demonstrate that taking an architectural perspective on the data ecosystem of a contact tracing app is valuable to understanding the complex and dynamic area of COVID-19 spread mitigation by means of apps, and the many fuzzy and interconnected causalities.

The architecture concept has been applied as a bridge concept between academic disciplines [9], e.g., to understand the impact of software architectures and organization on software engineering performance [8] or the effect of product architecture on innovation and market evolution [2]. Core to the architecture concept is the abstraction of the underlying detailed structural elements and behavioral properties of a complex system [29]. The abstraction reduces complexity for providing transparency about those complex systems and for supporting decisions about a system's design and evolution [19]. One common way of abstracting socio-technical elements and relations relevant to a phenomenon is conceptual modeling. In this paper, we apply conceptual modeling in the techno-legal context of COVID-19 contact tracing apps based on a data ecosystem architecture meta-model [7]. As will be described later, the modeling process will prove fruitful in terms of bridging the discursive gaps between the disciplines of IS and law.

While the architectural perspective has already been applied to complex socio-technical phenomena such as enterprise architecture [32, 34] we posit that this perspective can be further scaled up to account for data ecosystems. A data ecosystem is "a loose set of interacting actors that directly or indirectly consume, produce, or provide data and other related resources (e.g., software, services, and infrastructure). Each actor performs one or more roles and is connected to other actors through relationships, in such a way that actors by collaborating and competing with each other promote data ecosystems" [25]. These ecosystems are characterized by the openness, emergence, and dynamics of actors and relations [13].

Taking an architectural perspective allows a systematic decomposition of data ecosystems and provides transparency at different levels of granularity, including relations between actors involved (e.g., data flows, contracts, privacy statements), intra-organizational data processing (e.g., artificial intelligence (AI) with algorithms and purposes such as profiling or manipulation), and infrastructure (e.g., devices, applications, technical interfaces) [7]. In IS research, the architecture concept is used to capture and understand complex systems at different levels of granularity [32, 34]. Taking an architectural perspective on the data ecosystems of contact tracing apps can afford a clearer view of which decisions are especially sensible and quick to entail positive impact (e.g., higher efficiency, lesser invasiveness for privacy) or negative impact (e.g., the curtailment of fundamental human rights like privacy or freedom of assembly or movement).

The architectural perspective offers a socio-technical method for analyzing specific design decisions and their impacts at both the micro and macro levels. We envision that an interdisciplinary architectural approach will become conducive to interlinking a normative discourse and engineering decisions for key digital infrastructures of our societies. If so, such a method might be useful to policy makers interested in assessing design choices, either retroactively or proactively.

Our study began with questions concerning the function and usefulness of an architecture-driven analysis for inquiry and insights into normative issues. Our presupposition was that an architectural perspective establishes a fruitful interdisciplinary research tool at the interface of IS and law. Hence, our research question was: *Can the architectural perspective offer a suitable framework for analyzing the complex data ecosystems behind COVID-19 contact tracing apps?* By disclosing data flows and the interconnection of different stakeholders, the architectural perspective underlies design choices and facilitates the legal scrutiny of technical features.

The remainder of this paper is structured as follows. In the next section, we summarize literature on interdisciplinary research at the intersection of IS and law. Then, we outline our research approach, including a description of the architectural perspective and our multiple case study on contact tracing apps, especially by using the example of the Israeli HaMagen app. Next, we present our results, which consist of three different levels of new knowledge. Finally, we draw a conclusion by reflecting on our approach and findings.

2 RELATED RESEARCH

Scholars have already argued that bridging the gap between legal and technological approaches is necessary when measuring the sufficiency of new technological tools in satisfying legal requirements [24]. An early contribution was made by Herbert Burkert, who noted that privacy-enhancing technologies attempt to talk to system designers "in their own language," which is "one of abstraction followed by formalization" [6]. At the time, Burkert considered this approach attractive, but the deep gaps between lawyers and designers turned out nonetheless to be barriers to implement new technological approaches [4]. To date, legal scholarship that aimed explicitly at bridging these gap has focused mainly on the field of information privacy. In recent years, there has been a growing body of interdisciplinary literature on fairness and discrimination in machine learning (ML). As Nissim and Wood noted in the context of privacy, the reason for this may be

that due to a growing reliance on technological approaches, "privacy is increasingly becoming a technical concept, in addition to being a normative concept" [23].

Common to these attempts is the recognition that there is a gap between the disciplines and an understanding of the nature of this gap, along with the need to narrow it. For example, Nissim et al. made their contribution in the field of differential privacy, stressing that there are "significant gaps between legal and mathematical approaches to defining privacy" [24]. Their argument "bridges the gap between the privacy requirements of FERPA, a federal law that protects the privacy of education records in the United States, and differential privacy, a formal mathematical model of privacy" [24]. While mathematical models must make clear determinations, the legal vocabulary regarding the relevant concepts happens to be vague and thus cannot provide clear guidance for technological implementation [22, 24]. However, as Nissim and Wood argued, concepts such as privacy have a hybrid nature, meaning that they are a "multi-dimensional combination" of the legal and the technical, and thus understanding this hybrid nature "is necessary to bridge the gaps between the current legal and technical understanding of privacy" [23]. Nissim et al. also argued that legal and computer science concepts such as privacy are evolving simultaneously and co-dependently [24]. Therefore, both sides will benefit from a mutual dialogue. Moreover, even normative concepts, in this case privacy, vary in different cultures, and there are gaps between these concepts and their various implementations in laws and regulations [24].

Another contribution in the context of privacy, made by Birnhack et al., focused on the difficulties of implementing the concept of privacy by design (PbD) and suggested that discursive gaps between the legal field and the field of engineering can explain such difficulties [4]. Their findings were that "PbD seems an intuitive and sensible policy tool for lawyers; however, for information systems developers and engineers, it is anything but intuitive as it goes against the grain of several well-established principles of information systems engineering" [4]. Their article unfolds and explains the technological challenge of translating normative concepts into concrete technical requirements. The authors argued that when the subject matter of legal regulation is technology, then "despite the neutral language, the law cannot be oblivious to its subject matter" [4], meaning that the law has to acquire an adequate perception of how the regulated activity operates or how it should operate. One way of achieving that requirement, Birnhack et al. suggested, is that "the law should engage with technology, and lawmakers should converse with designers" [4].

However, this previous literature looks past the ecosystem phenomenon. Therefore, we argue that taking an architectural perspective is needed, since it is much broader than approaches that focus on one specific law or technology. For example, in the context of ML in content moderation, the focus of public debate and legal inquiry is often on the algorithm itself (e.g., how it was trained), and not enough on how it is shaped by the ecosystem (e.g., strategic players that could affect the learning outcome).

A different contribution takes a novel approach to educating both law and technology students. Hod et al. argue that "an effective multidisciplinary dialogue between data scientists and lawyers is needed" [12] in the field of AI, in order to ensure that "AI systems are 'responsible' and incorporate important social values such as fairness, accountability and privacy" [12]. A joint dialogue would enable both lawyers and computer scientists to "spot relevant technical, ethical and legal issues and to jointly reach better outcomes early in the design stage of the system" [12]. For achieving that goal, the researchers have designed a course called Responsible AI, Law, Ethics and Society, which engages joint teams of data science and law students in joint problem-solving tasks on real-world AI challenges, thus developing an effective multidisciplinary dialogue.

There have also been previous approaches to combining law and technology from the IS field, particularly architecture and modeling research. For example, Becker et al. developed a meta-design to integrate regulatory requirements into collaborative IS [3]. This meta-design includes a design layer, an implementation layer, and a model-instance-bridge. The design layer contains modeling techniques and tools for the design of, e.g., lawful processes or collaboration structures, while the implementation layer embodies the realized instance of a designed model. The model-instance-bridge ensures consistency between these two layers by proposing approaches such as model-driven software development. Siena et al. introduced a goal-oriented modeling technique to capture legal requirements and integrate them into IS design and development [28]. Other authors have suggested modeling methods for legal risk management to proactively improve contractual decisions [18].

From a legal perspective, the importance of architecture and modeling entered the discourse as early as the 1990s, when legal scholars introduced the theoretical framework of regulation by design. Most notable was Joel R. Reidenberg's work on "Lex Informatica," which pointed to the interrelation between technological affordances and human behavior [27]. Another influential work was Lawrence Lessig's key model for conceptualizing the codependency of major regulating forces of human behavior: law, social norms, markets, and code [17]. Lessig's model thus added algorithms to the forces governing human behavior, coining the phrase "code is law" [17]. Since then, there has been a stable stream of legal scholarship aimed at refining our understanding of the mutual relations between technology and human behavior. Our approach provides new insights by addressing the interactions between the different actors participating in an ecosystem and by zooming in on a particular design in order to identify specific legal issues. Furthermore, our focus is not on exploring the normative quality of technology, but on describing complex ecosystems to answer normative questions.

Knackstedt et al. suggested conceptual modeling in law [16]. They argued that different approaches to conceptual modeling stemming from the IS field should be integrated into the legal domain to enhance the comprehensibility of legal regulations and contracts. In their paper, they summarized six requirements for legal communication and suggested a corresponding research

agenda for conceptual modeling in law. This research agenda proposes that both the creation and application of law can be supported through the modeling aspects of method generation, model generation, model application, and model evaluation. The authors also underlined that "meta modeling could provide the basis for legal conceptual modeling approaches" [16].

Summing up, the existing literature highlights the need for interdisciplinary research between IS and law, especially with regard to privacy. On this topic, the literature mentions some challenges to be considered (e.g., the need for joint dialogue) and proposes a few approaches for interdisciplinarity (e.g., Becker et al.'s meta-design [3]). However, the literature falls short of supporting interdisciplinary research based on an architectural perspective that takes into account the complex phenomena occurring in contemporary data ecosystems, and that suggests related socio-technical modeling elements to enable a dialogue between IS and legal scholars.

3 RESEARCH APPROACH

3.1 Introducing the Architectural Perspective

In our research, we applied an architectural perspective to study the complex data ecosystems of COVID-19 contact tracing apps. Architecture is generally defined as "the fundamental concepts or properties of a system in its environment embodied in its elements, relationships, and in the principles of its design and evolution" [15]. On the one hand, we were motivated by the wish to experiment with a novel form of architecture-driven interdisciplinary research at the interface of IS and law, and on the other hand, we aimed at exploring the privacy issues and policy implications that accompany contact tracing apps. While there is a large body of knowledge of the manifold data storage approaches and data processing mechanisms of these apps, we argue that capturing the data ecosystem architecture of these apps provides a more comprehensive understanding, because not only specific technologies are considered, but also the different actors involved, their interfaces and data flows, as well as underlying influences such as regulation mechanisms. Moreover, taking an architectural perspective allows researchers to identify blind spots and open legal questions, as well as potential risks for human rights such as privacy harms.

To capture and analyze contact tracing apps, we used the data ecosystem architecture meta-model according to Burmeister et al. [7] (see appendix). A meta-model is defined as "the description of the set of notations and concepts used to define a model" [14]. According to Frank, meta-models foster a common language among stakeholders and indicate "what to model" by proposing key elements and relations from a holistic point of view [10].

In the literature, meta-models on data ecosystems are scarce. However, by contrast with the approaches described in the previous section, the meta-model from Burmeister et al. suggests concrete socio-technical elements and relations to be considered in the context of data ecosystems [7], thereby providing us with the opportunity for detailed modeling of contact tracing apps. Moreover, while for example the meta-design from Becker et al. assumes separated layers for design and implementation [3], using a meta-model in our research approach is more integrative because it serves as a formal modeling template. The meta-model from Burmeister et al. also originates from the privacy field, which is relevant to our context [7].

Figure 1 outlines our research approach. We studied and modeled contact tracing apps, as real-world phenomena, by referring to the elements and relations proposed by the meta-model. More specifically, we instantiated the meta-model to get concrete models of the contact tracing apps we studied. The exemplary model in Figure 1 illustrates an extract of modeling HaMagen, the Israeli contact tracing app. Both IS and legal scholars reflected on the created models to derive novel insights, leading to a reflux of knowledge regarding both the meta-model and the phenomena, as well as interdisciplinary research in general. For example, we made some adjustments to the selected meta-model to entirely capture specifics of contact tracing apps, and we identified privacy issues and open questions in the context of these apps.

The meta-model proposed by Burmeister et al. consists of five horizontal layers (data ecosystem, business, legal, data, and IT landscape layers), with each proposing specific modeling elements [7]. For example, the legal layer comprises elements such as consent or privacy statement, and the IT landscape layer covers applications or technical interfaces. In addition, the meta-model suggests three vertical layers (individuals' architecture, boundary architecture, and enterprise architecture) that assign the modeling elements to specific actors and their collaboration [7]. Here, the individuals' architecture refers to elements (e.g., consent, device) pertaining to individuals (e.g., app users), and the enterprise architecture represents elements of organizations, including business processes or databases. Finally, the boundary architecture includes elements at the interface of two actors, i.e., shared or jointly used elements such as services, data objects, or privacy statements. We used the meta-model from Burmeister et al. as a template for iteratively modeling the different contact tracing apps we studied and as deductive guidance on what to look for in the data ecosystems of these apps [7]. The following sections describe our process of working with this meta-model in detail, thus affording the reader an impression of how this theoretical explanation translates to practice.

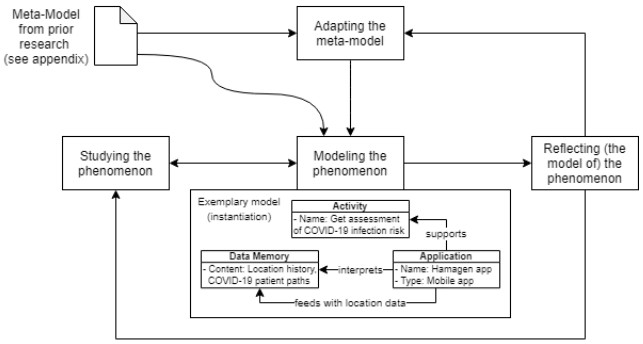

Figure 1: Research approach

3.2 Multiple Case Study

We conducted a multiple case study [33] of different COVID-19 tracing apps developed in Germany and Israel to understand the implications of different design choices regarding COVID-19 digital infrastructures. Applying the meta-model to these apps supported our assessment of the impacts of technology-related policy choices, during crisis and beyond, including their impact on human rights.

Israel offered an interesting case study, as its government simultaneously applied two parallel strategies to tackle the outbreak phases of the pandemic. The first strategy was the use of the national security surveillance infrastructure of the Israeli General Security Service (GSS) in order to trace the locations of confirmed COVID-19 patients. The use of the centralized architecture of the GSS surveillance was operated covertly and was mandatory. The second strategy was a voluntary open-source contact tracing app named HaMagen ("The Shield"), developed and launched by the Israeli Ministry of Health (MoH). Its system architecture was mostly decentralized, and location data were stored on the users' mobile phones and checked against a central database of confirmed COVID-19 patients administered by the MoH.

Germany, on the other hand, offered informative cases that were very different in their approach and development from the Israeli ones. One example from Germany is the Corona-Warn-App, a decentralized governmental architecture combined with an interface offered by Apple and Google and installed on users' smartphones. The second German example is the privately owned and developed contact tracing app Luca, which gave us the possibility of studying a data ecosystem not established by governmental actors.

Although we studied all four cases using the meta-model, in this paper we focus on the HaMagen app. It should be stressed that although the meta-model could facilitate a systematic comparison of these apps, we left such a comparison outside the scope of this paper. We believe that focusing on one example is worthwhile, because it enables us to detail the modeling process in full and to delve into the complexity of the relevant data ecosystem, thus demonstrating the characteristics of the interdisciplinary dialogue. Since this was an exploratory research, we deemed a comparison of different contact tracing apps to be beyond the scope of this paper. Moreover, focusing on one example more clearly exhibits the usefulness of the modeling for lawyers, regulators, and civil society advocates by demonstrating in detail the functionality of the meta-model.

3.3 Case Study of HaMagen ("The Shield") App

3.3.1 Background and Description. HaMagen was launched by the Israeli MoH on March 22, 2020. The app's main function was to inform users if and when they had been exposed to a positive-testing COVID-19 patient. Its original version was based on GPS technology and collected location data, whereas the app's update included both GPS and Bluetooth technologies. The app's use was voluntary, as it required the user's authorization to access the tracking services of the device, and the collected data was kept on the user's device only. No information was transferred from the app to the MoH unless infected users voluntarily consented to share their data with the MoH. The app's code was open-source.

A prestigious team of civil experts voluntarily advised the MoH on several aspects of the app's design. The app was later tested by multiple cyber and data security agencies, including staff of the Israel National Cyber Directorate, specialists from the commercial sector, and leading experts from the civil cyber and data security community in Israel. The testing included open source code checks against the privacy policy published by the MoH [22]. The app's design and performance were largely supported and mainly praised by professionals. Thus, the state of Israel had, from the beginning of the pandemic, a state-of-the-art piece of contact tracing technology. The app's assimilation prospects appeared promising.

At first, the Israeli public seemed keen on using the HaMagen app, and a week after its launch the app had about 1,000,000 users. However, its success was temporary, as was the case in other states. In the first two months after its launch, the app was downloaded 2,254,085 times. From that point onward, users began deleting the app from their devices. By July 27, 2020, when HaMagen 2 (the updated version) was launched, 1,175,257 users had already deleted the app. At the end of November 2020, the MoH announced that there would be no further investment in the app. We further analyze the failure of the app's assimilation in section 4.1.

3.3.2 Data Collection and Analysis. Since there is so far no established method for conducting interdisciplinary architecture-driven research, we followed an explorative research approach. In the following, we describe the process of our inquiry using the HaMagen app as an example. Although this is a report on only one case study, we believe that this example can serve as a necessary first step in the direction of establishing a new method for interdisciplinary research.

We conducted our case study on HaMagen in three iterations. In each iteration, we collected and analyzed different types of data, performed a corresponding stage of modeling, and discussed our findings. Key questions that led our inquiry were, for example, who are the stakeholders involved in the data flow? Which actors exert influence? What infrastructure do they use? What are the privacy issues that arise?

We started our first iteration by searching for and collecting formal technical documents about the HaMagen app. The data material comprised descriptions provided by authorities [20], process models found on the web [22], as well as the GitHub documentation [11]. We then conducted an in-depth analysis of this material. First, we compared the collected documents by looking for deviations. For example, across the process models, we identified divergent steps describing the app's functioning. Second, we marked elements and relations mentioned in the material and matched them with the meta-model. For example, "location history" was characterized as "data object" or "Ministry of Health" as "actor." Based on these findings, we conducted our first stage of modeling.

The result of the first stage is presented in Figure 2 and is referred to as the primitive model. This initial phase of the model raised many open questions for both IS scholars and legal experts involved in the research project. Therefore, in the second iteration, we refined our primitive model by collecting and analyzing further data material. This material consisted of legal documents, such as HaMagen's privacy policy [21], and scientific literature on HaMagen. However, we found that there are only few papers explaining HaMagen's functioning. Moreover, these papers either focus on specific mechanisms (e.g., key derivation) [26] or describe the app's architecture in a rather isolated and basic form [30, 31], i.e., without details of the app's data ecosystem and similar to the public documents collected in our first iteration. Nevertheless, our analysis of HaMagen's privacy policy, in particular, led to new insights in this second iteration. We marked elements and relations that our primitive model had not taken into consideration. For example, the privacy policy mentioned specific types of data or third parties. This resulted in several improvements in the form of an advanced model within a second stage of modeling, presented in Figure 3.

While we assumed the privacy policy would have elucidated HaMagen's data ecosystem, discussing our advanced model in a third iteration led to further essential questions from the perspectives of both IS scholars and lawyers. Consequently, we conducted five informal interviews with experts involved with the app, two in the structuring of the MoH's database and three in the operation of the human epidemiological investigation unit. The questions asked during the interviews referred mainly to overall data processing in the app's data ecosystem, its sources of location data, and the involvement of third parties. Participants did not wish to be recorded, therefore only notes were taken. The novel insights generated by the interviews resulted in a third stage of modeling, illustrated in Figure 4.

During each iteration, multiple project meetings took place (approximately once every two weeks between 2021 and 2022), wherein the researchers (three IS scholars and three legal experts) discussed the findings and the created models. This resulted in a mutual learning process, because the IS scholars reflected on legal questions and documents and refined the architectural models, while the lawyers provided legal and normative explanations when possible, searched for the requested additional materials, and suggested improvements for the meta-model and the modeling procedure.

3.3.3 Stages of Modeling.

Stage 0: The Motivation for Applying Ecosystem Architecture Modeling. Generally, the IS scholars and the lawyers shared a common motivation, i.e., to improve the dialogue with the other discipline in order to gain and refine an understanding of the operations and effects of technology, and of COVID-19 contact tracing apps in particular. Moreover, both groups shared a motivation to inform the meta-level of the interdisciplinary dialogue, i.e., to contribute to understanding the barriers to and gaps in such mutual inquiry and to find ways of overcoming them. These motivations are rooted in a shared understanding of the mutual influence and co-dependency of normative and legal systems and socio-technical systems.

Indeed, from the lawyers' perspective, an initial motivation was to experiment with a possible mode of an unfamiliar interdisciplinary dialogue, in the form of architectural models. From the IS scholars' perspective, an initial motivation was the need to apply an existing theoretical meta-model on real-world cases. In the following, we unfold our three stages of modeling.

Stage 1: The primitive model. In our initial modeling attempts, we tried to understand how the app was working based on formal technical documentation. This information was the basis for creating a primitive version of the model (see Figure 2). For simplification purposes, the model only includes a selective legend (a complete description of all the modeling elements can be found in Burmeister et al. [7]).

The first presentation of the model raised a discussion between the IS scholars and the legal experts in the group as to several of the app's features. The IS scholars were asking, for example, whether users downloaded the app voluntarily or how user consent was acquired, as these issues were not explicit in the technical documentation. Although such questions were easy for the lawyers to answer, there were further details that the IS scholars felt were missing, and that challenged the legal experts' knowledge of the app and its operation. The main questions here were focused on the MoH's database (symbolized by the question mark in Figure 2): Where does the information about COVID-19 patients come from? What are the data sources of the COVID-19 patient paths? How are these paths generated? In short, the lawyers were asked about the structure of the data flows and of the database, questions that were meaningful in terms of users' rights. Another question was about how exactly the HaMagen app creates the location data. To what extent is the operating system involved? The lawyers were then sent to look for information that they initially had not considered necessary. These further investigations were the basis for the creation of the second version of the model.

It should be noted that this primitive phase of modeling already demonstrates how architecture modeling provides a common ground for discussion among group members. It also stresses the significance of the visualization as a way of bypassing the limits of our own disciplinary concepts and as a means for collaboratively spotting new issues and identifying gaps in our knowledge of the examined system.

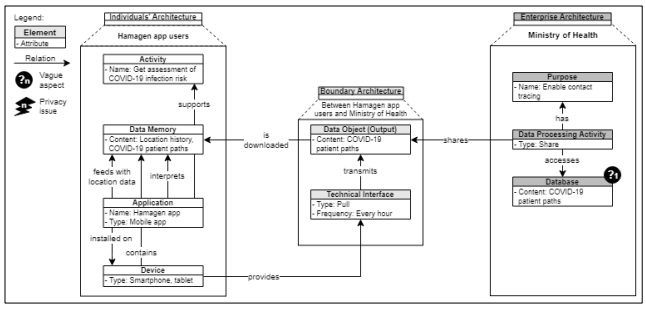

Figure 2: Primitive model of HaMagen data ecosystem

Stage 2: The advanced model. The next stage was based on privacy policy documentation and on published scientific literature. It should be noted that we did not conduct a code review against the privacy policy, since that had already been done by local experts and supported by the MoH [22], as was already mentioned in section 3.3.1.

Among the new information delivered to the IS scholars was information about laboratories and the epidemiological investigation unit, the existence of the MoH's epidemiological system, and the involvement of Microsoft Azure as the MoH's cloud services provider. Figure 3 shows the updated version of the model that the IS scholars created. Here, from a legal point of view, many important questions arose with regard to third-party implementation. For example, potential consequences for the app's data ecosystem must be considered, e.g., if Microsoft changes contractual terms or the cloud service is disrupted.

The primitive stage of modeling led to the discovery of more actors involved in the app's data ecosystem, such as cloud service providers. Although our starting point was privacy-centric, the second stage of modeling based on this discovery revealed a data ecosystem much bigger and more complex than we had initially assumed. Similarly to the primitive model, vague aspects or open questions were symbolized by question marks. Moreover, referring to the meta-model, we also highlighted some conspicuous privacy-critical issues to be considered (symbolized by the flashes in Figure 3).

The new model was much more nuanced and it demonstrated the intricacies of the app's data ecosystem. We noticed some issues worth mentioning in terms of privacy. Issues 1a and 1b relate to the data obtained, kept, and processed by the MoH. In HaMagen's privacy policy, it is written that user information labeled as "irrelevant" within the epidemiological investigation process is stored. However, issue 1a clarifies that only users' location histories and proximity data are transmitted from the HaMagen app to the MoH, and only in cases of infected users who voluntarily consented to this. Therefore, it is unclear what "irrelevant" information means according to the privacy policy and how it is received and processed. Moreover, the privacy policy states that the location histories of users and "irrelevant" information are kept for a duration of at least seven years, resulting in privacy concerns that need to be addressed (issue 1b). With regard to the purposes of contact tracing and infection control, it is incomprehensible why users' location histories are kept for seven years, and it is highly questionable why the "irrelevant" information is stored at all, being, as it is, according to HaMagen's privacy policy, irrelevant to the epidemiological investigation process. Finally, issue 2 highlights that the COVID-19 patient paths are uploaded on the Microsoft Azure cloud service, which distributes the data to the HaMagen app. This leads to some open questions with regard to the anonymization of the data and the extent to which Microsoft can access the data, e.g., for administration or other purposes.

The researchers identified several additional open questions based on the advanced model. The first question relates to some ambiguity regarding user consent (first question mark in Figure 1). The app's privacy policy states that verified patients who consent to sharing their data can decide whether their location history and proximity data are to be shared jointly or separately. However, this option is not clarified and from an architectural perspective it is questionable whether or not users' location histories and proximity data would be stored on separate databases depending on the users' choices. The second question relates to the WiFi history that is kept by the app (second question mark). The privacy policy only indicates that the WiFi history is stored in the data memory of the users' devices, but does not describe the corresponding purpose or whether the WiFi history is used for cross-referencing. However, the most striking thing was that instead of making the picture clearer, the advanced model was a manifestation of our shortage of understanding regarding the operation of the MoH's database and its implications for users' rights. Furthermore, the analyzed legal materials, e.g., the privacy policy, did not clarify what exactly the epidemiological system is (third question mark) and how it extracts the patient paths from the MoH's database.

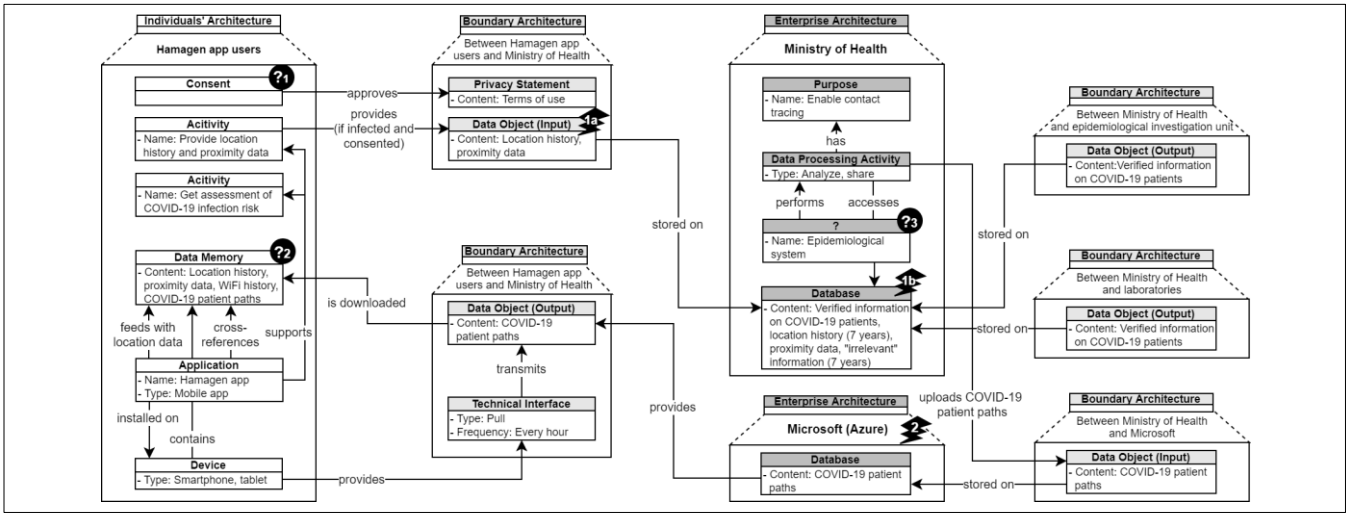

Figure 3: Advanced model of HaMagen data ecosystem

The main conclusion at this point was that the best direction for further inquiry would be to look into the identities, the functions, and the actions of extra actors in the app's ecosystem. The mid-stage of modeling signifies the fact that our perception was limited not only by our conceptual framing, but also by the legal materials that we could study. It turned out that these materials do not always reveal everything; the legal experts learned that they did not have enough material to understand the real essence of the data ecosystem under investigation.

Stage 3: The final model. The final version of the model (see Figure 4) was created after additional actors had been investigated with regard to the MoH's database. This investigation was based on five interviews with Israeli experts who were collecting essential information about data processing within the MoH and was followed by filing an open information request to the MoH together with the Israeli privacy NGO Privacy Israel. The two important actors that were added to our model at this stage were the GSS, which is a governmental actor, and Systematics, which is a private provider of geographic IS. Moreover, information about input and output data was added as part of the epidemiological investigation unit's operations. Similarly to the previous model, the MoH's epidemiological system that is responsible for data processing remained a black box to us (first question mark in Figure 4). Furthermore, the final model left us with even bigger questions than our initial ones with regard to data flows, data sources, data holders, and data controllers: What kinds of data can Systematics access (second question mark)? What specific data are inputs for and outputs of the epidemiological investigation unit (third question mark)? To what extent is the GSS involved in the data ecosystem of HaMagen (fourth question mark)? At the time of writing these sentences, our information request to the MoH has not been answered. We followed procedure and filed a complaint to the government concerning this administrative failure in accountability. Since our complaint was not answered either, we are about to appeal, together with Privacy Israel, to the Israeli High Court of Justice.

4 RESULTS

The results of this study can be conceptualized as belonging to different levels of knowledge. The first level is the actual knowledge of the exemplary contact tracing app we studied (HaMagen) and its data ecosystem; the second level is knowledge of the meta-model we used, its benefits and its shortcomings; and the third level is knowledge of the interdisciplinary research process of acquiring common knowledge shared by IS scholars and legal experts.

4.1 Knowledge of the Data Ecosystem: Novel Insights into HaMagen

The first level of results proved our interdisciplinary research approach and the meta-model fruitful in terms of acquiring knowledge of the specific data ecosystem studied, i.e., HaMagen. Our main contribution to understanding the HaMagen app and similar systems is that the question of privacy can only be fully answered from a data ecosystem perspective. In our case study of HaMagen, we learned that although the app was privacy-friendly for its users, it was nonetheless unfriendly for COVID-19 patients in that their personal information was needed for the app's proper functioning. This knowledge is important not only as a historical analysis of HaMagen and its ecosystem, but also for the future of similar ecosystems and their development and deployment. It is already evident that the infrastructure that was created to support contact tracing during the COVID-19 crisis is here to stay; therefore, understanding the political and socio-technical challenges of such ecosystems is relevant to developers, policy makers, advocates, and other stakeholders.

In addition, we believe that the meta-model also contributes to the understanding of another issue concerning the app's operation, namely the question of the need for a civil app in the face of the state's use of the GSS surveillance. The final model demonstrates the possibility of two different scenarios: in the first scenario, data from GSS surveillance and from the human epidemiological investigation unit is involved in the operation of

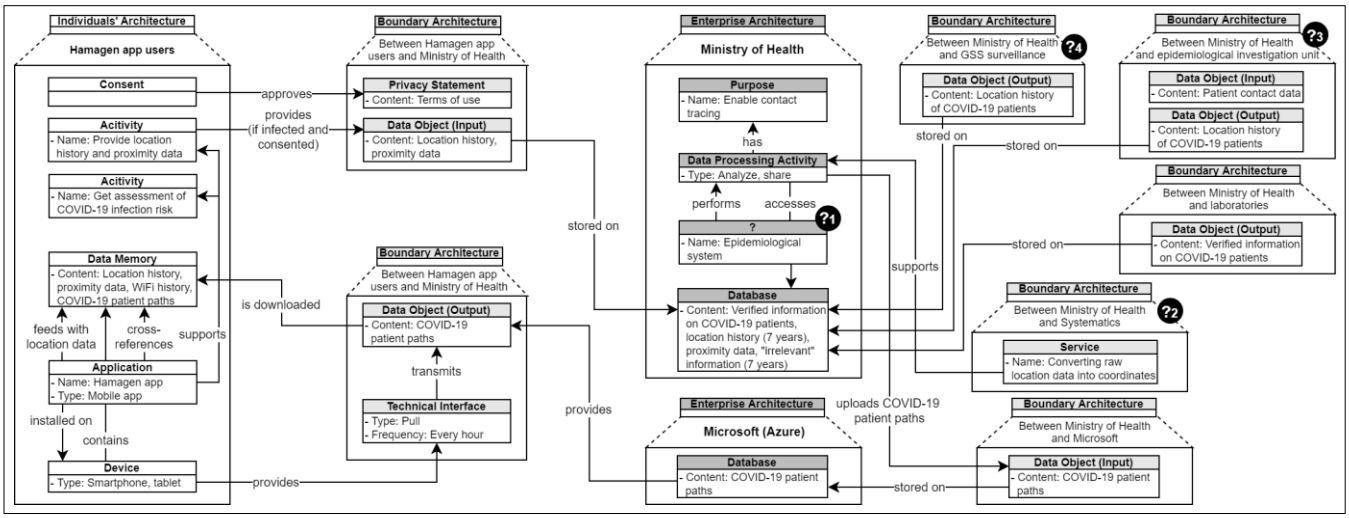

Figure 4: Final model of HaMagen data ecosystem

HaMagen and is correlated. If this scenario is to be proven the right one, then the civil app's users were deceived as to the sources of the information that was fed into the app; in this case, the critique claiming that the Israeli citizenry rejected the app because of the government's use of GSS surveillance is plausible [31]. In the second scenario, only the human epidemiological investigation unit is used as a source for the app's data flows. Therefore, only partial location data based on interviews or protocols is transferred to the app, since full data must contain precise location data that is received from the GSS. This scenario results in a partial, inaccurate operation of the app. If this scenario is to be proven the right one, then the app's users were misled as to the app's functionality and were getting a crippled product. Although we currently lack evidence that can determine which scenario is more plausible, we nonetheless believe that each of the scenarios opens a path for explaining HaMagen's failure to be assimilated.

4.2 Knowledge of the Architectural Meta-Model

The second level of results proved our interdisciplinary research approach fruitful in terms of acquiring knowledge of the architectural meta-model that was used. Applying the meta-model from Burmeister et al. to modeling contact tracing apps led to some interesting insights regarding the meta-model's utility [7]. In addition, some shortcomings throughout the three modeling stages became clear.

The first modeling stage demonstrates the shortcomings of a model that is mainly process-based and may be similar to a conservative IS perspective, i.e., where the focus is often on understanding the sequence of business processes. Here, we instantiated the meta-model based on existing process models and public information. The primitive model (see Figure 2) simply describes the rough functionality of the contact tracing app, but lacks many important details from a legal perspective. For example, the primitive model implies that the app may be privacy-friendly to its users, but obscures the origin of COVID-19 patient paths as the key data involved, as well as the potential privacy implications for affected patients.

The second modeling stage demonstrates the shortcomings of instantiating a meta-model based mainly on legal documents such as privacy policies. This stage adds the process that may be similar to a conservative legal perspective. At this stage, our advanced model (see Figure 3) proves that, at least in HaMagen's case, privacy statements are not entirely effective or sufficient for identifying who is involved in the complex data ecosystem of a particular application. Using privacy statements as the main data source for modeling data ecosystems may not provide a complete picture of the actors collecting, holding, processing, or sharing data, but rather shows only the tip of the iceberg. For example, at this stage, it remained unclear how exactly the verified information about COVID-19 patients is turned into concrete location data and whether or not a third-party data processor is involved.

Finally, the third modeling stage demonstrates that something that appeared naive during the first modeling stage turned out to be the most important open legal issue. In our example of the HaMagen app, this was the MoH's database. While this database was added as a modeling element in the first stage, its nature became less clear as we expanded our investigation across the stages. For example, after studying the app's privacy policy, the database's content attribute changed and the epidemiological system of the MoH, which processes the database, was added. However, the third stage shows that even after interviews with involved experts, it still remains unclear how exactly the COVID-19 patient paths are determined and how the epidemiological system works.

Our case study also led to some suggestions for improving the elements and relations included in the meta-model. First, the legal experts remarked that highlighting the direction of data would make the instantiations of the meta-model, i.e., the concrete architectural models, more understandable. Therefore, we added input and output labels to the data objects. We also included an indication of the duration of data storage in the database element, which is particularly important to discussing privacy issues. Furthermore, we adjusted the labels of some relations in our models to clarify certain issues. For example, we indicated that the COVID-19 patient paths, representing data objects, are actively downloaded by the HaMagen app. Here, we believe that the meta-model is somewhat rigid and too formal in terms of defining relations between modeling elements. Instead, a more flexible and pragmatic specification of relations during the modeling process may be useful to illustrate circumstances and dependencies in data ecosystems more clearly. Most notably, while the applied meta-model only suggests marking specific privacy issues, we added question marks in our architectural models. We believe that this is particularly important to highlighting vague elements or open questions, which, as demonstrated in our example of HaMagen, may lead to further iterations of research and related stages of modeling.

Demonstrating the application of the meta-model through contact tracing apps also highlights the meta-model's usefulness and advantages. While the elements and relations suggested by the meta-model not only help to summarize what happens in a specific data ecosystem, they also support the identification of obfuscated parts or open legal questions, e.g., when information about specific elements cannot be found. Thus, we argue that the architectural meta-model has several notable methodological contributions. By accumulating issues in a processual manner, it has not only a synchronic dimension, i.e., the capturing of a system's relevant elements at one concrete point in time, but also a diachronic dimension as well. The linear diachronic dimension captures different states of the studied data ecosystem, thus monitoring the ecosystem's evolution. Additionally, it captures different stages of the researchers' knowledge, thus manifesting the evolution of knowledge.

Applying the meta-model can also support societal decision-making. The HaMagen case is an example of an ex-post analysis of the studied system and it exemplifies the utility of the meta-model as a mode of explanation and of scrutiny. However, since the meta-model allows us to easily model and compare to-be scenarios, we believe it to be an efficient tool for ex-ante decision

making as well. For example, the meta-model and its derivatives can be used as a basis for creating better privacy policies and for assessing the impact of using novel technological solutions.

4.3 Knowledge of the Interdisciplinary Research Process

The third level of results proved our approach fruitful in terms of acquiring knowledge of interdisciplinary research in general. Despite the fact that this knowledge stems from a single project's process analysis and the study of one specific type of data ecosystems, i.e., contact tracing apps, we believe that at least some of our results may be generalized and can thus be valuable to other scholars interested in learning about complex socio-technical ecosystems. Moreover, we believe that our approach can provide some guidance and tools for other interdisciplinary research projects for several reasons.

First, applying an architectural perspective supports a common reality for both IS scholars and lawyers based on a shared terminology that is realized through a meta-model and its instantiations. Conducting such an architecture-driven approach in interdisciplinary research can support the construction of reality in a way that all members of a research project can look at and understand. Although each discipline retains its own "version" of reality, architectural models can create a link between the respective reality constructions. Through such a link, scholars from different disciplines can contribute their individual perspectives in a collaborative fashion and gain interdisciplinary and novel insights into the phenomena they are exploring. We argue that taking an architectural perspective based on a meta-model may be the closest approximation of a common language, as the meta-model can serve as a medium for communication and ground interdisciplinary discussion.

Second, architecture modeling supports gaining transparency about and in-depth understanding of the complexity of a specific phenomenon in a systematic manner. Here, the modularity of a meta-model helps researchers model parts of a phenomenon (e.g., a data ecosystem) step by step and gradually arrive at a whole picture. Moreover, as we demonstrated in our example of the HaMagen app, each step may lead to new information requirements for the model's refinement toward addressing the open questions identified. In this sense, scholars from different disciplines may function as each other's "instructors," e.g., the lawyers collect information that IS scholars use to create the architectural models, while the IS scholars explain their methods and understandings to the lawyers.

Third, our research approach proved the existing literature's claim as to a main difference between the discourses to be valid. While IS scholars require accurate information to model complex systems, legal scholars are much more familiar with vagueness and with the possibility of a problem to be answered with more than one binary answer. Actually, the concept of binary decisions is not completely unfamiliar to law, e.g., many legal procedures end with the assessment of an act as lawful or unlawful. However, since natural language and social processes are involved in legal interpretation and decision-making, lawyers are accustomed to the vagueness inherent in all human activity. To better understand how the other discipline "thinks," we not only had to become familiar with that difference; in fact, we realize that the meaning of a better understanding was that while the lawyers in the group had to look for more details than they are used to, the IS scholars had to accept the fact that sometimes architectural elements will remain indeterminate.

5 CONCLUSIONS

In this paper, we described and analyzed the process and results of our joint interdisciplinary research project. The project applied an architectural perspective to study the complex data ecosystems behind COVID-19 contact tracing apps. We argued that combining architectural and normative perspectives is a novel interdisciplinary research approach that bridges the disciplines of IS research and law.

While this paper is descriptive and mapped an exploratory interdisciplinary research process, it demonstrates that taking an architectural perspective on complex socio-technical ecosystems is valuable for interdisciplinary research between IS and law. Referring to and using architectural models helped us structure the debate and initiate a co-learning process between IS scholars and lawyers. This may provide an important basis for developing an interdisciplinary method by systematically integrating the legal perspective in both the development of a meta-model and the derivation of models. We can already see that this method will generate a process of collaborative knowledge production and we also have an indication that architectural models support structuring this process. However, further research is required to substantiate this process at the level of more general theories of knowledge creation so as to build strong interfaces between disciplinary methods.

We are aware of several limitations. First, the research is so far exploratory and rests on one analytical case. However, the results provide a proof of concept that following an architectural approach facilitates the interdisciplinary creation of knowledge and provides avenues for future research. This future research can further corroborate the generalizability of the architectural approach. Second, there is always the possibility that a different meta-model will render different results. However, the selected meta-model proved fruitful in ways that surpassed our initial expectations. As a result, until we become aware of a better conduit for this interdisciplinary effort, research on the further adaptation and extension of the meta-model is warranted.

We conceptualized the results of our joint inquiry as belonging to three distinguished levels of knowledge. The first level included knowledge of the specific data ecosystem we studied, namely HaMagen. The second level included knowledge of the meta-model we used, its merits and shortcomings, and the solutions that were selected in the process to improve the meta-model and adjust it for mutual dialogue. The third level included knowledge of the interdisciplinary research approach itself, and here we used a critical, reflexive mode of inquiry into the merits and shortcomings of our own methods to improve general and specific

understanding of attempts to create a common language for acquiring shared knowledge. We also believe that our interdisciplinary approach has several practical implications. First, it can provide the basis for an impact assessment, be it by the service provider, a regulator, or a third party. The mapping of actors, data flows, and boundary architectures are especially relevant to assessing possible risks to (legally protected) interests like privacy. In the same way, an evaluation can benefit from the information offered by an architectural model of the ecosystem. This function is not only needed for the aforementioned policy-making purposes, but also for regulatory oversight, e.g., as a way of detecting compliance, and for allocating risks or liabilities created by participants. Furthermore, the model can help to draft more meaningful privacy policies. It shows what potential users need to know before they consent to data processing. Finally, policy makers can use such models to identify design options and entry points for intervention.

The many possible outlooks that we imagine demonstrate the fertility of the chosen meta-model and research approach. Architectural modeling may also prove useful in analyzing social problems that involve technology. For example, we could use the architectural perspective to studying data ecosystems in other socio-technical contexts, such as conversational bots designed to automate communication.

Reflecting on our project and its findings led to a general recognition of their contribution to the political understanding of socio-technical ecosystems and their effects. The architectural perspective demonstrates a critical legitimacy crisis — in governments, legal systems, and society in general — created by the transformation from the traditional (social) way of producing norms into a technological production of norms. Law is a social practice. If we lose the legitimate basis upon which social and legal norms stand, then the conditions under which the law is produced vary. We believe that a first necessary step in addressing such pressing political issues is making them understandable to all interested stakeholders.

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REFERENCES

[1] Nadeem Ahmed, Regio A. Michelin, Wanlin Xue, Sushmita Ruj, Robert Malaney, Salil S. Kanhere, Aruna Seneviratne, Wen Hu, Helge Janicke, and Sanjay Jha. 2020. A Survey of COVID-19 Contact Tracing Apps. *IEEE Access* 8, 134577-134601.

[2] Carliss Y. Baldwin and Kim B. Clark. 2000. *Design rules: The power of modularity.* MIT Press, Cambridge, MA, USA.

[3] Jörg Becker, Marcel Heddier, Sebastian Bräuer, and Ralf Knackstedt. 2014. Integrating Regulatory Requirements into Information Systems Design and Implementation. In *Proceedings of the 35th International Conference on Information Systems*, Auckland, New Zealand.

[4] Michael Birnhack, Eran Toch, and Irit Hadar. 2014. Privacy Mindset, Technological Mindset. *Jurimetrics* 55, 1 (2014), 55-114.

[5] Isobel Braithwaite, Thomas Callender, Miriam Bullock, and Robert W. Aldridge. 2020. Automated and partly automated contact tracing: a systematic review to inform the control of COVID-19. *The Lancet Digital Health* 2, 11 (2020), 607-621.

[6] Herbert Burkert. 1997. Privacy-Enhancing Technologies: Typology, Critique, Vision. In *Technology and privacy: the new landscape*, P. E. Agre and M. Rotenberg (eds.). MIT Press, Cambridge, MA, USA, 125-142.

[7] Fabian Burmeister, Christian Kurtz, Pascal Vogel, Paul Drews, and Ingrid Schirmer. 2021. Unraveling Privacy Concerns in Complex Data Ecosystems with Architectural Thinking. In *Proceedings of the 42nd International Conference on Information Systems*, Austin, TX, USA.

[8] Marcelo Cataldo and James D. Herbsleb. 2012. Coordination breakdowns and their impact on development productivity and software failures. *IEEE Transactions on Software Engineering* 39, 3 (2012), 343-360.

[9] Regine Dörbecker, Daniela Böhm, and Tilo Böhmann. 2015. Measuring Modularity and Related Effects for Services, Products, Networks, and Software - A Comparative Literature Review and a Research Agenda for Service Modularity. In *Proceedings of the 48th Hawaii International Conference on System Sciences*, Kauai, HI, USA.

[10] Ulrich Frank. 2014. Multi-perspective enterprise modeling: foundational concepts, prospects and future research challenges. *Software & Systems Modeling* 13, 3 (2014), 941-962.

[11] GitHub. 2022. Hamagen, https://github.com/MohGovIL/hamagen-react-native, last accessed February 24 2022.

[12] Shlomi Hod, Karni Chagal-Feferkorn, Niva Elkin-Koren, and Avigdor Gal. 2022. Data Science Meets Law: Learning Responsible AI together. *Communications of the ACM* 65, 2 (2022), 35-39.

[13] Marco Iansiti and Roy Levien. 2004. *The Keystone Advantage: What the New Dynamics of Business Ecosystems Mean for Strategy, Innovation, and Sustainability.* Harvard Business School Press, Boston, MA, USA.

[14] ISO/IEC. 2002. Information technology - CDIF framework - Part 1: Overview. Standard 15474-1:2002.

[15] ISO/IEC/IEEE. 2011. Systems and software engineering – architecture description. Standard 42010:2011.

[16] Ralf Knackstedt, Marcel Heddier, and Jörg Becker. 2014. Conceptual Modeling in Law: An Interdisciplinary Research Agenda. *Communications of the Association for Information Systems* 34, 36 (2014), 711-736.

[17] Lawrence Lessig. 2006. *Code: Version 2.0.* Basic Books, New York, NY, USA.

[18] Tobias Mahler. 2009. Visualising Legal Risk. Jusletter IT.

[19] Nenad Medvidovic and Richard N. Taylor. 2010. Software architecture: foundations, theory, and practice. In *Proceedings of the ACM/IEEE 32nd International Conference on Software Engineering*, Cape Town, South Africa, 471-472.

[20] Ministry of Health. 2022a. HaMagen 2.0, https://govextra.gov.il/ministry-of-health/hamagen-app/download-en/, last accessed February 24 2022.

[21] Ministry of Health. 2022b. Privacy Policy and Information Security, HaMagen Application, https://govextra.gov.il/ministry-of-health/hamagen-app/magen-privacy-en/, last accessed February 24 2022.

[22] Omri S. Moyal. 2020. "Hamagen" Application - Fighting the Corona Virus, https://medium.com/proferosec-osm/hamagen-application-fighiting-the-corona-virus-4ecf55eb4f7c, last accessed February 24 2022.

[23] Kobbi Nissim and Alexandra Wood. 2018. Is Privacy Privacy? *Philosophical Transactions of the Royal Society A* 376, 2128 (2018).

[24] Kobbi Nissim, Aaron Bembenek, Alexandra Wood, Mark Bun, Marco Gaboardi, Urs Gasser, David R. O'Brien, Thomas Steinke, and Salil Vadhan. 2018. Bridging the Gap between Computer Science and Legal Approaches to Privacy. *Harvard Journal of Law & Technology* 31, 2 (2018), 689-780.

[25] Marcelo I. S. Oliveira, Glória de Fátima Barros Lima, and Bernadette F. Lóscio. 2019. Investigations into Data Ecosystems: A Systematic Mapping Study. *Knowledge and Information Systems* 61, 2 (2019), 589-630.

[26] Benny Pinkas and Eyal Ronen. 2021. Hashomer – Privacy-Preserving Bluetooth Based Contact Tracing Scheme for Hamagen. In *Workshop on Secure IT Technologies against COVID-19.*

[27] Joel R. Reidenberg. 1998. Lex Informatica: The Formulation of Information Policy Rules Through Technology. *Texas Law Review* 76, 3 (1998), 553-594.

[28] Alberto Siena, John Mylopoulos, Anna Perini, and Angelo Susi. 2009. Designing Law-Compliant Software Requirements. In *Proceedings of the International Conference on Conceptual Modeling*, Gramado, Brazil, 472-486.

[29] Herbert A. Simon. 1962. The Architecture of Complexity. *Proceedings of the American Philosophical Society* 106, 6 (1962), 467-482.

[30] Ruoxi Sun, Wei Wang, Minhui Xue, Gareth Tyson, Seyit Camtepe, and Damith C. Ranasinghe. 2021. An Empirical Assessment of Global COVID-19 Contact Tracing Applications. In *Proceedings of the 43rd International Conference on Software Engineering*, Madrid, Spain, 1085-1097.

[31] Eran Toch and Oshrat Ayalon. 2021. How mass surveillance can crowd out installations of COVID-19 contact tracing apps.

[32] Robert Winter and Ronny Fischer. 2007. Essential Layers, Artifacts, and Dependencies of Enterprise Architecture. *Journal of Enterprise Architecture* 3, 2 (2007), 7-18.

[33] Robert K. Yin. 2009. *Case Study Research: Design and Methods.* Sage, Thousand Oaks, CA, USA.

[34] John A. Zachman. 1997. Enterprise Architecture: The Issue of the Century. *Database Programming and Design* 10, 3 (1997), 44-53.

APPENDIX

This appendix shows the meta-model used in our research. See Burmeister et al. [7] for more details.

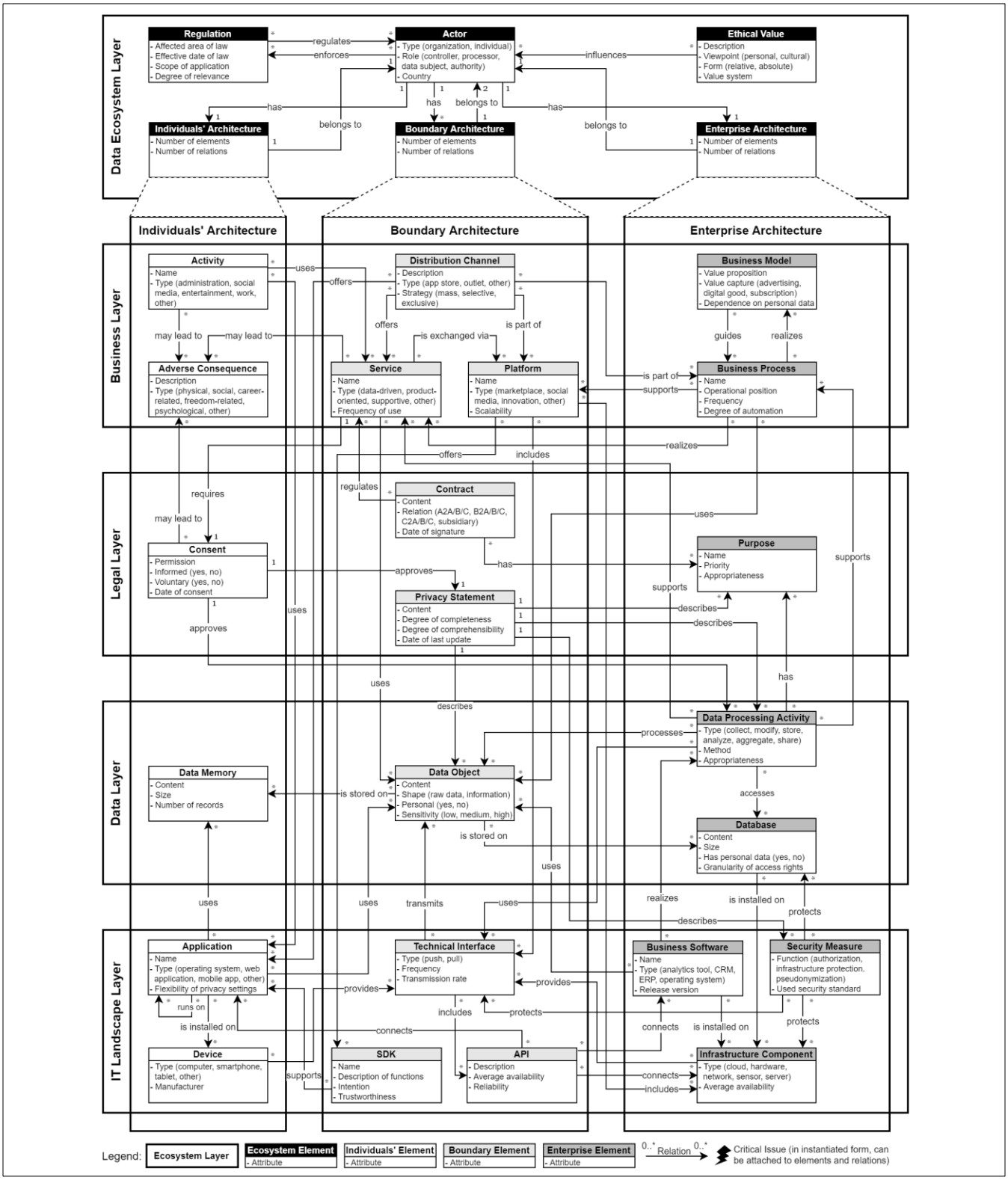

Figure 5: Data ecosystem architecture meta-model [7]